

Stopwatch Feature Report

Keypad-driven stopwatch for the TM4C123 light program

This report describes a stopwatch feature added to the TM4C123GH6PM LaunchPad traffic-light program. The stopwatch is controlled entirely by three keys on the EduARM4 keypad, wired as genuine hardware interrupts.

1. Objective

Extend the existing LED blink-rate/colour controller (blinking an RGB LED at a chosen speed and colour, shown on a 4-digit 7-segment display and controllable by on-board switches or UART) with a stopwatch: a timer that can be enabled, started, stopped, paused, and resumed, displaying elapsed minutes and seconds on the same shared display.

2. Software Design

The stopwatch is modeled as a four-state machine:

- DISABLED — the feature is off; the shared display shows the LED readout instead.
- IDLE — enabled, elapsed time at zero, not counting.
- RUNNING — actively counting up.
- PAUSED — counting frozen, elapsed time retained.

Three keys drive transitions between these states:

Key	Effect
"1"	Toggles between DISABLED and IDLE (enabling always resets elapsed time to zero).
"2"	From IDLE, starts running. From RUNNING or PAUSED, stops completely and resets to zero.
"3"	From RUNNING, freezes the count. From PAUSED, resumes counting from where it froze.

Each key's full transition logic lives directly inside the GPIO Port C interrupt handler, since the keypad is the only input this feature responds to.

3. Timing

A SysTick interrupt fires every 10 milliseconds. It increments elapsed time (in hundredths of a second) whenever the state machine is RUNNING, and it also serves as the timestamp source for debouncing the three keys: any second edge on the same key within 200 ms of the previous one is ignored as switch bounce.

4. Display Behavior

The 4-digit 7-segment display is shared between two purposes. Ordinarily it shows the LED controller's state: a fixed letter, the current blink rate, the current colour, and a running/paused indicator. The instant the stopwatch leaves the DISABLED state, the same four digits switch to showing elapsed time as MM:SS, and they switch back the moment it returns to DISABLED.

5. Conclusion

This design adds stopwatch functionality using genuine interrupt-driven input, isolated cleanly from the rest of the program: it shares the display and the SysTick timer with the existing LED logic, but introduces no new dependency on any external library, and required no changes to the project's startup file.